NUMERICAL INVESTIGATION OF A SELF-SUSTAINING THERMALLY COUPLED AMMONIA DECOMPOSER–COMBUSTOR SYSTEM

By
Thomas Abraham
(22JE1018)

Objectives:

- Establish a **validated ammonia decomposer baseline** based on Chein et al.
- Develop a **scaling strategy** to increase mass flow while preserving plug-flow behavior
- Design a **hydrogen–air combustor** to generate high-temperature flue gases
- Develop an **annular decomposer configuration** enabling thermal coupling via recirculation
- Analyze **heat transfer and recirculation requirements** using NTU-based methodology
- Implement and evaluate a **fully coupled CFD model**, addressing key numerical challenges

Numerical modeling of hydrogen production from ammonia decomposition for fuel cell applications

Chein et al.

- **MSR studies** (methanol steam reforming) had already established a well-tested CFD/plug-flow framework for **porous catalyst beds**.
- **Chellappa et al. (2002)** provided experimental kinetics for NH_3 decomposition on **Ni–Pt/Al_2O_3**, showing that the reaction is **first-order in NH_3 concentration** with **negligible H_2 inhibition at ~1 bar**.
- This justified using a **simplified Arrhenius rate law** of the same form as those employed in MSR models.
- **Chein's group (2010)** applied this framework with Chellappa's kinetic constants to build a **1-D plug flow model** that:
    - matched experimental results well, and
    - rivaled the accuracy of more expensive 2-D CFD simulations.
- The **1-D simplification** was justified because:
    - Catalyst bed **porosity and particle diameter** had negligible effect on cracking efficiency, and
    - The reactor was small in radial dimension, so **temperature, pressure, and velocity gradients** across the radius could be neglected.

Set-Up

Equations solved for the mathematical modelling of the combustor will take the following assumptions into account:

1. The flow is axisymmetric, laminar and steady.
2. Ideal gas behaviour
3. The catalyst particles are assumed to be spherical with diameter 'd_p' and the catalyst bed is treated as a porous medium with homogeneous porosity 'ε' and permeability 'k'.
4. The catalyst bed is in local thermodynamic equilibrium with the surrounding gas.
5. Diffusion is neglected as Convection dominates (High Peclet Number)

Mass Conservation:

$\nabla \cdot (\rho \mathbf{V}) = 0$

Momentum Conservation (Brinkman–Darcy–Forchheimer Model):

$(1/\varepsilon^2)\, \nabla \cdot (\rho \vec{V}\vec{V}) = -\nabla p + (1/\varepsilon)\, \nabla \cdot (\mu \nabla \vec{V}) - (\mu/K)\vec{V} - \rho C_f \sqrt{(1/K)}\, |\vec{V}|\vec{V}$

Energy Conservation:

$$\nabla \cdot \left(\varepsilon \rho c_p \vec{V} T\right) = \nabla \cdot \left(k_{eff}\, \nabla T\right) + \left(\frac{\varepsilon^2 \mu}{K}\right) |V|^2 + q_c$$

Species Transport Equation (for species 'i'):

$\nabla \cdot (\varepsilon V C_i) = \nabla \cdot (D_i \nabla C_i) + R_i$

Simulation Type:
2D **Axisymmetric** model representing the cylindrical packed-bed reactor used by Chein et al. (2010).
Isothermal steady-state simulation (energy equation not solved after patching the domain isothermally to 893K).
Conducted using **ANSYS Fluent (Pressure-Based Solver).**

Models and Physics:
Flow model: Laminar, steady, pressure-based formulation.
Pressure–velocity coupling: SIMPLE algorithm
Spatial discretization: Second-order upwind for momentum and species

Porous media: Darcy–Forchheimer model applied through user-defined coefficients:

$$K = \frac{d_p^2 \varepsilon^3}{150\left(1-\varepsilon\right)^2}, C_F = \frac{1.75}{\sqrt{150}\varepsilon^{3/2}}$$

Species transport: Enabled for NH_3, H_2, and N_2 with a **volumetric first-order Arrhenius reaction**:

$$r_R = k_R C_{NH_3}$$

where $k_R = C_R\ (1\ -\ \varepsilon)\rho_s A\ R\ T \cdot\ exp(-E\ /\ R\ T)$

Boundary Conditions:

Velocity Inlet:

$T_{in} = 893\ K$,NH_3 mass fraction = 1.0

$u = u_{in}$,computed from $W/F = 10$

Pressure Outlet:

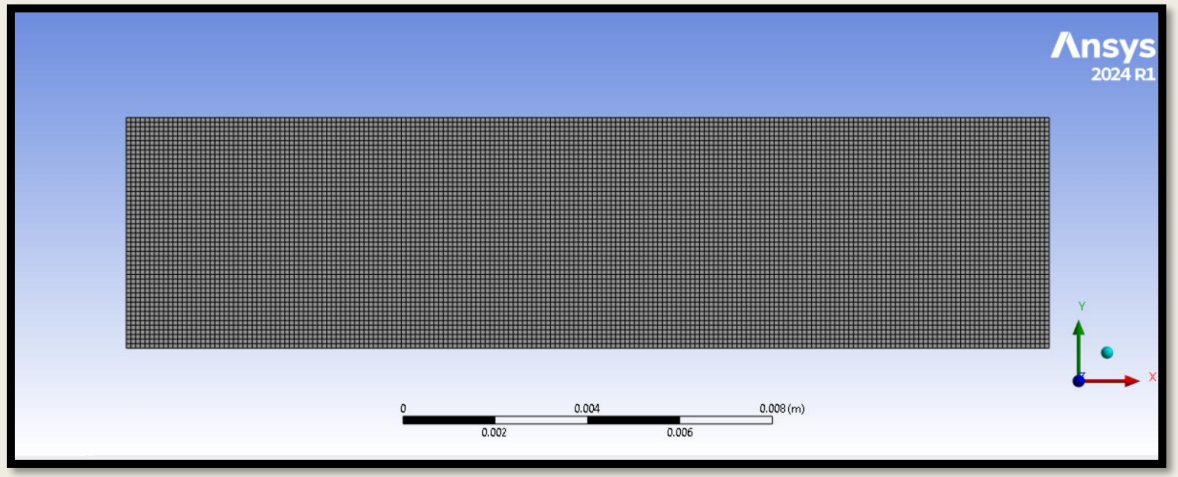
$p_{out} = 1\ bar$

Backflow total temperature = $893\ K$

Wall: No-slip condition $(u = v = 0)$,adiabatic, $\left(\frac{\partial T}{\partial r} = 0\right)$

Axis: Axisymmetric condition $\left(\frac{\partial u}{\partial r} = \frac{\partial T}{\partial r} = \frac{\partial C_i}{\partial r} = 0\right)$

The temperature equation is not being solved-
The domain is isothermally maintained at 893K.



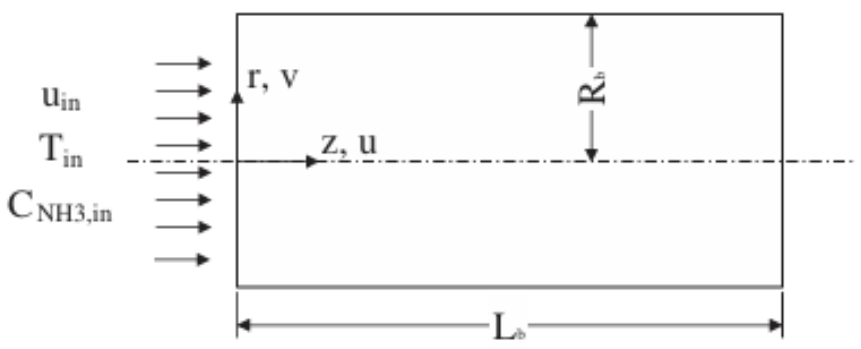
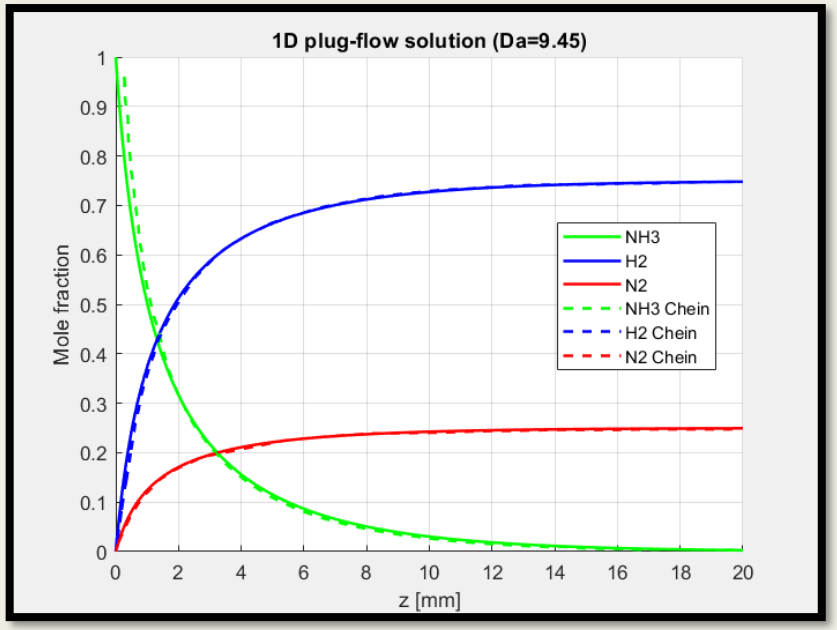


Fig. 1 – Schematic of NH_3 decomposition reactor physical domain.

Mesh:
Face meshing with quadrilateral elements, edge sizing done along axial and radial directions such that the a uniform grid with an aspect ration of ‘1’ is generated (for anisotropic resolution)

Validating the Chemical Kinetics and the Porous Media Physics

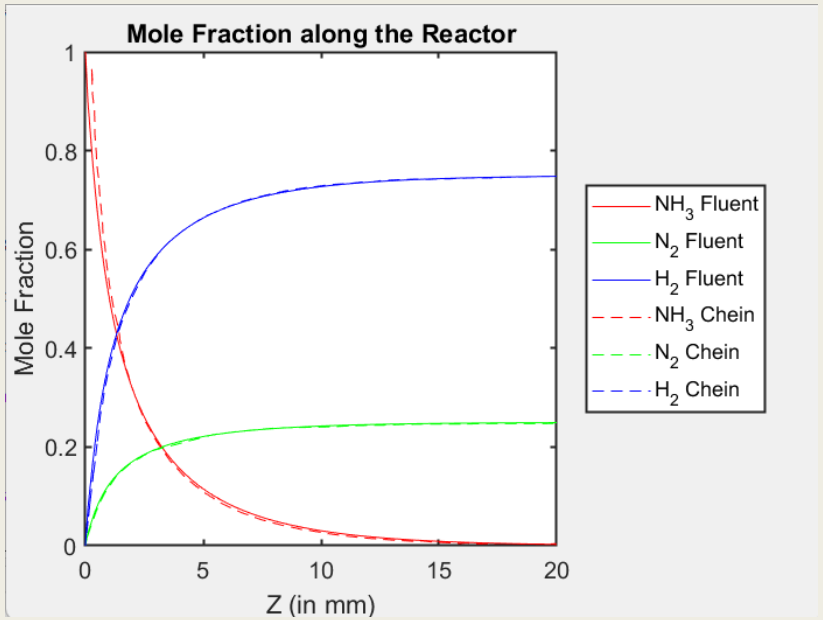


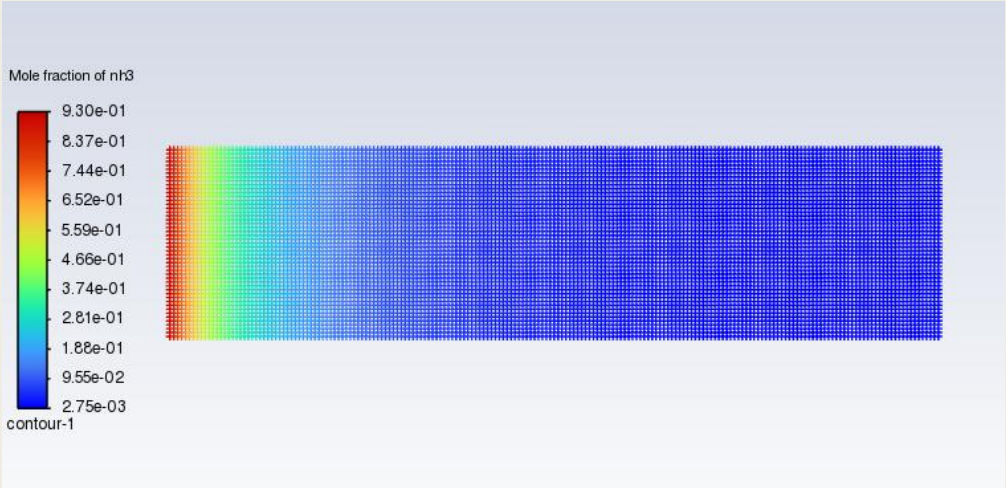
- Reaction kinetics validated independently using the MATLAB 1D plug-flow model based on Chein's derived relation: $z \,/\, (u_{in} \,/\, k_r) \;=\; h \;-\; 2 \cdot ln(1 \;-\; h)$
- Porous-medium behavior verified using Ergun's pressure-drop formulation:

$$\Delta \mathrm{P} = -\left((\mu/K)v + \left(\rho C_F/\sqrt{K}\right)|v|v\right) * \mathrm{L}$$

Quantity	MATLAB formulation from Chein et al.	Fluent (Area-Weighted Average)	Error
Pressure Drop over the Length of the Reactor (20mm)	662.64 Pa	664.38 Pa	0.2626%

Results



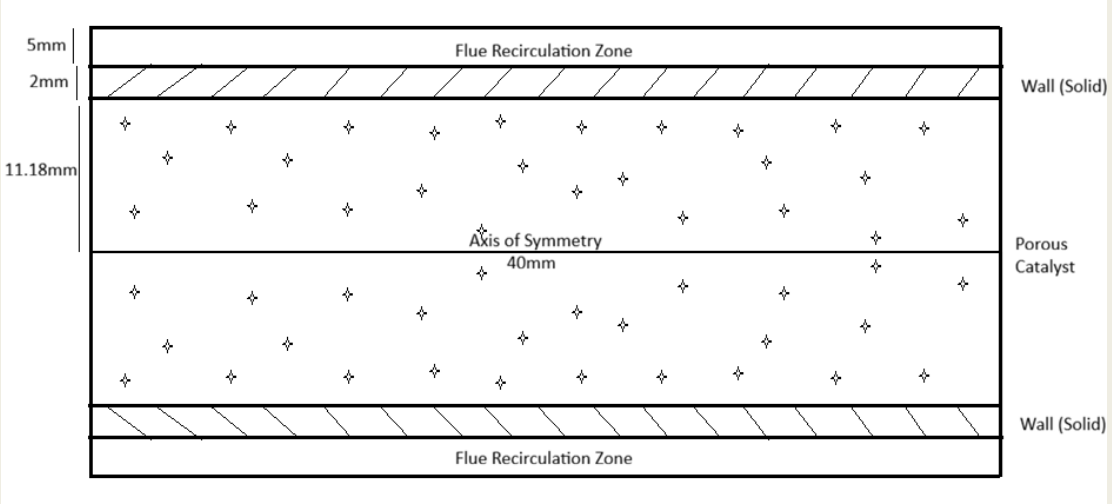


The predicted ammonia conversion at the reactor outlet is approximately 99.72%, corresponding to an outlet mole fraction of 0.00276. This is in close agreement with the near-complete conversion reported by Chein et al. under similar operating conditions.

Redesigned Decomposer:

Redesigned Decomposer
Baseline decomposer mass flow rate **insufficient** for combustor operation
Further downscaling of combustor **not feasible** → redesign required

Scaling Strategy
Reactor radius scaled by $\sqrt{5}$ → **5× increase in mass flow rate**
Inlet velocity kept **constant → residence time preserved**
Catalyst properties and kinetics **unchanged**



Physical Justification
Reactor operates under **isothermal, plug-flow conditions**
Negligible radial gradients
Cross-sectional scaling **does not alter reaction physics**

Annular Extension
Decomposer reconfigured into an **annular geometry**
Inner region: **porous catalytic bed (NH_3 decomposition)**
Outer region: **recirculated hot flue gases**
Enables **thermal coupling without external heating**

Decomposer Design

$$\dot{Q}_{req} = \dot{m}_{NH_3}\left(\Delta H_{rxn} + C_p(T_{dom} - T_{amb})\right)$$
$$\dot{\mathrm{Q}}_{\mathrm{avail}} = \dot{m}_{flue} C_{p\ \mathrm{flue}}(T_{flue} - T_{dom})$$

$$\varphi = \frac{\dot{\mathrm{Q}}_{\mathrm{req}}}{\dot{\mathrm{Q}}_{\mathrm{avail}}}$$

Chein's reactor maintains an isothermal domain at 893K. Thus, $T_{dom} = 893\mathrm{K}$.
From the simulation the values of '$C_{p\ \mathrm{flue}}$', species composition and $T_{flue} = 2506.5829\mathrm{K}$
The calculated value of the recirculation ratio (φ) was- 0.2416.

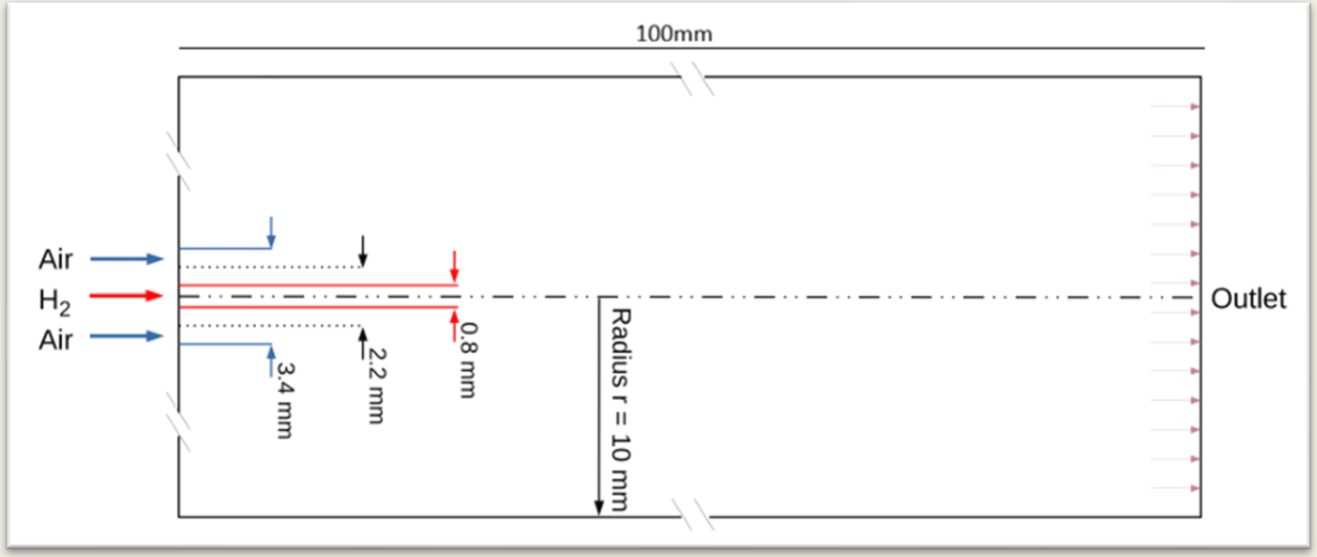
Since only the inlet temperatures are known, the effectiveness of the heat exchanger could be characterised by the "Number of Transfer Units".

$$\mathrm{NTU} = \frac{\mathrm{UA}}{\dot{\mathrm{m}}\mathrm{C_p}}$$

The calculated NTU for this set-up: 34.4755
Since $\mathrm{NTU} \gg 1$, The decomposer temperature is not limited by heat-transfer resistance and the required temperature level (893 K) is controlled primarily by energy balance ($\dot{m}$ and φ) rather than geometric UA constraints.

Combustor Design

- Annular oxidizer inlet surrounding a central hydrogen fuel jet
- Promotes stable diffusion flame formation and efficient mixing



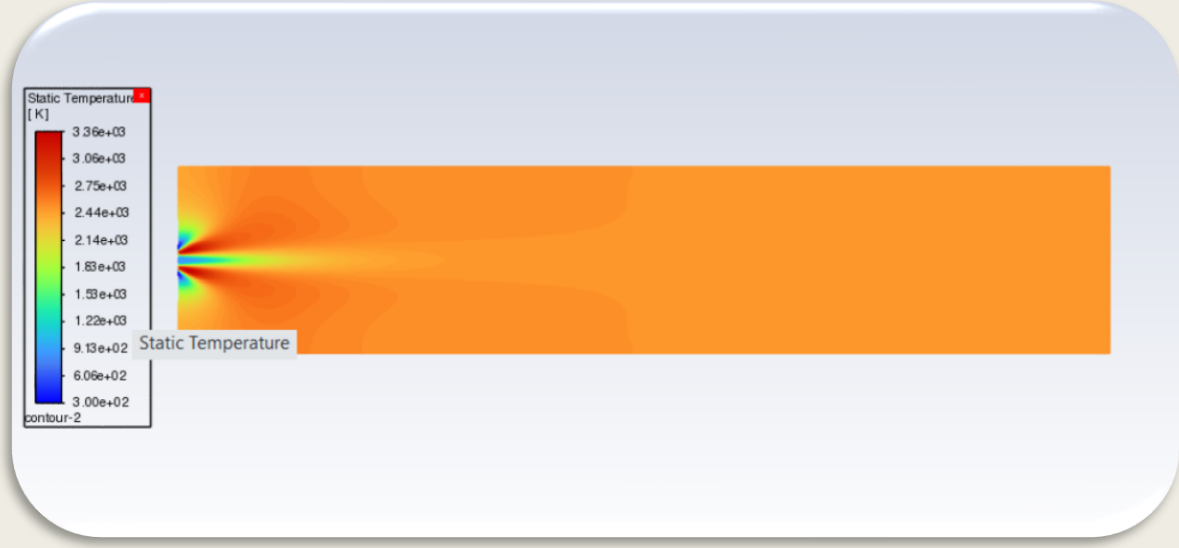
- Geometry inspired by literature (Barabás et al.), but axially extended
- Required due to higher mass flow rates in the present design
- Extension ensures the pressure outlet boundary is placed downstream
- Minimizes interaction with strong recirculation zones
- Prevents **backflow-induced instabilities** and improves convergence

Key Outcome:

- Achieved a numerically stable and physically consistent flame structure
- Suitable for integration with the decomposer via thermal coupling

Stoichiometric Combustion ($\lambda = 1$)

$mfr_{fuel} = 7.5692e - 07\ \mathrm{Kg}/sec$
$\mathrm{T_{fuel}} = 893K$
Species Composition-
$\mathrm{X_{H_2}} = 0.75, \mathrm{X_{N_2}} = 0.25$
$Velocity = 12.95793\ m/\mathrm{sec}$ (from Reference Values)



$mfr_{fuel} = 4.5978e - 06\mathrm{Kg/sec}$
$\mathrm{T_{fuel}} = 893K$ (ambient)
Species Composition-
$\mathrm{X_{O_2}} = 0.21, \mathrm{X_{N_2}} = 0.79$
$Velocity = 0.7433223m/\mathrm{sec}$ (from Reference Values)

Outlet Flue Conditions:
Species Composition-
$\mathrm{X_{H_2O}} = 0.309906, \mathrm{X_{N_2}} = 0.6892626, \mathrm{X_{H_2}} = 5.5010547e - 11, \mathrm{X_{O_2}} = 0.00083144166$
$\mathrm{T_{air}} = 2506.5829\mathrm{K}$
$mfr_{flue} = 5.3546611e - 06\ \mathrm{Kg/sec}$

MATLAB-Based Recirculation Calculation

Decomposer Inlet (NH_3 Stream)

$\dot{m}_{NH_3} = 7.56 \times 10^{-7}$kg/s

$T_{in} = 893$K

Pure NH_3 feed

Flue Gas Inlet (From Combustor)

Properties obtained via **area-weighted averaging**:

T_{flue} ,$\dot{m}_{flue}$

Species composition

$C_p,\ M,\ \rho$

MATLAB-Based Coupling

Combustor outputs used to perform **energy balance**

Recirculation ratio ϕcomputed

Required flue mass flow:

$$\dot{m}_{recirc} = \phi \cdot \dot{m}_{flue}$$

Implementation in CFD

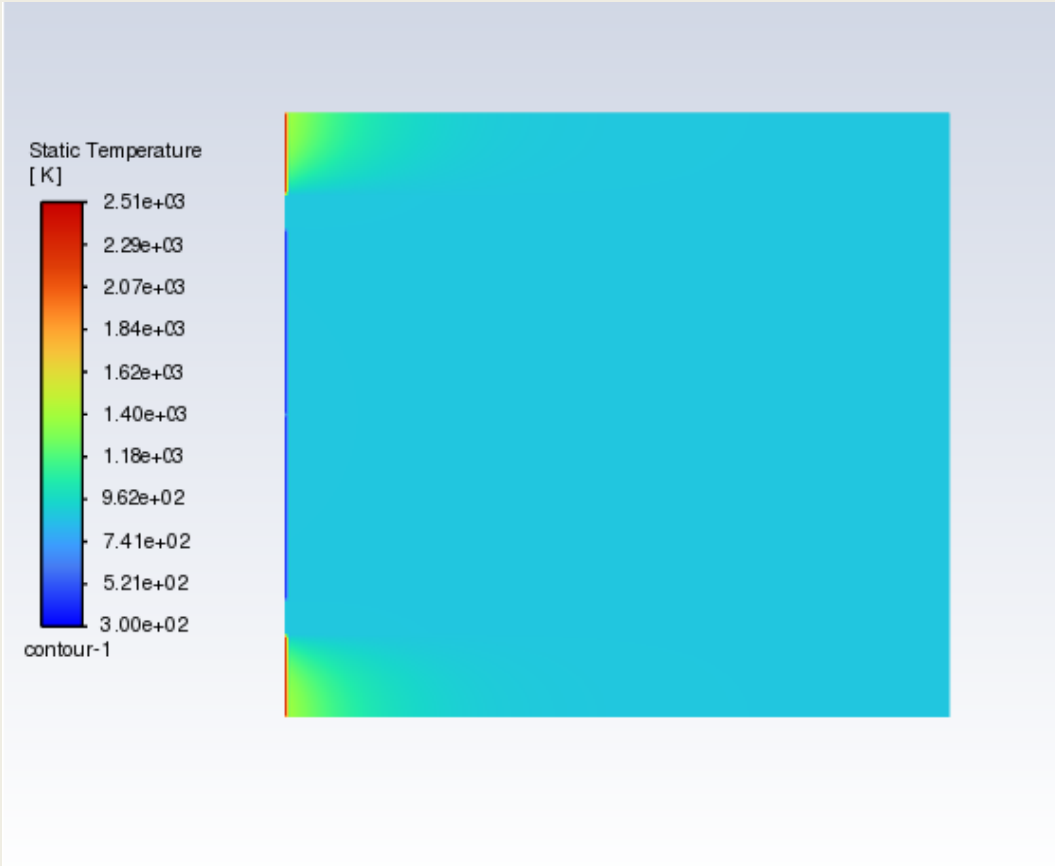
Recirculated flue supplied to **annular region**

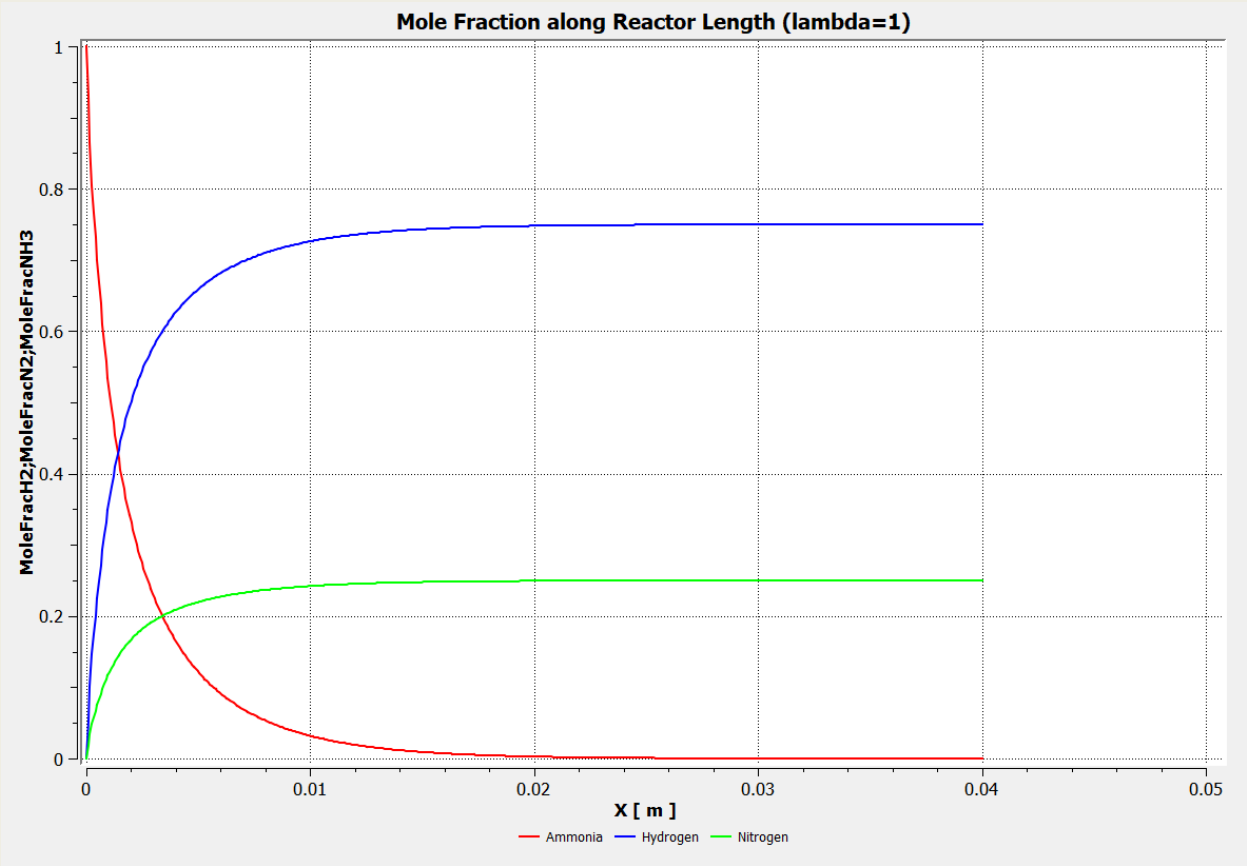
NH_3 supplied to **porous core**

Interface walls: **coupled heat transfer**

Key Role

Ensures correct thermal input to maintain $T_{decomposer} \approx 893\ K$

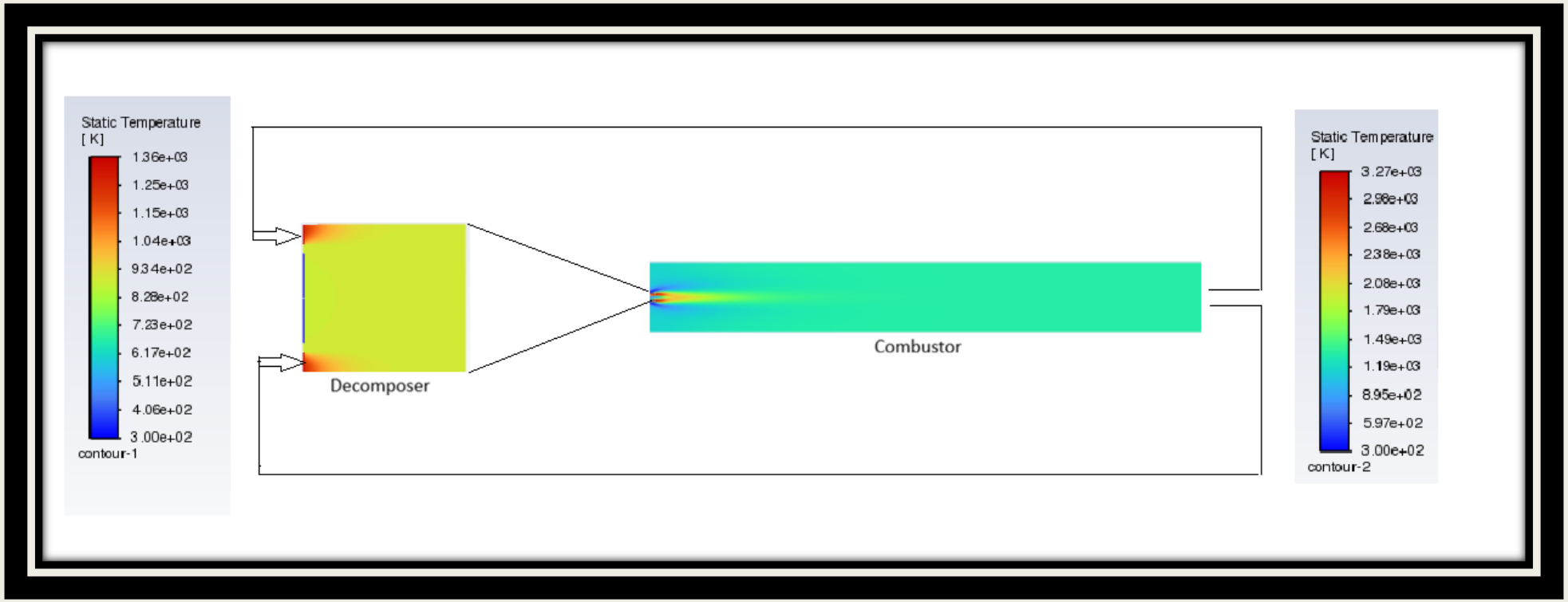




Recirculation ratio from MATLAB: 0.2416
Recirculation Ratio used fine-tuned from the simulation: 0.2635
Resulting interior temperature- $892.9934K \approx 893\mathrm{K}$

Thus the recirculation ratio from MATLAB and the actual simulation are in good agreement.

Similar analysis for lean combustion ($\lambda = 2.85$) yielded:



- Lean combustion (λ = 2.85)
- Flue temperature ≈ 1357 K
- The resulting $\mathrm{NTU} = 9.9055 \gg 1$
- $\varphi_{\mathrm{MATLAB}}$=0.4134
- φ_{CFD} =0.4329 (Porous catalyst temperature= 892.59867K)
- Higher ϕ required due to lower flue temperature
- MATLAB prediction remains consistent with CFD tuning
- System achieves stable thermal coupling.

The pressure outlet has been further moved axially to accommodate for the increased mass flow.

Zone-Specific Diffusivity Modeling Using UDFs

Physical / Numerical Observation

- Chein's decomposer behaves effectively as a **plug-flow reactor**.
- Standard Fluent diffusion treatment introduced significant **axial diffusion** inside the porous catalytic region.
- This produced **non-physical inlet gradients** in the species mole-fraction profiles.
- The resulting transport behavior resembled a **diffusion–reaction boundary value problem (BVP)** rather than the expected convection-dominated plug-flow system.

Evidence of Non-Physical Diffusion Effects

- Strong sharp gradients developed near the decomposer inlet when standard diffusivity models were used.
- Species profiles became diffusion-dominated and deviated from Chein et al.'s reported plug-flow behavior.
- The issue originates from Fluent solving the full convection–diffusion–reaction transport equation, whereas the physical reactor behavior is largely convection–reaction dominated.

Conflicting Transport Requirements

Porous Decomposer Region

•Requires negligible axial diffusion to preserve plug-flow behavior.

•Transport should be governed primarily by:

- convection
- finite-rate reaction source terms

Combustor Region

•Requires physically realistic species diffusion to sustain the hydrogen–air laminar diffusion flame.

•Diffusion is essential for:

- species mixing
- flame stabilization
- thermal transport

•Therefore, a single global diffusivity model was unsuitable for the coupled system.

Numerical Strategy: Zone-Specific Diffusivity

Decomposer Zone

•Species diffusivity artificially suppressed:

$$D \sim 10^{-20}\ m^2/s$$

•Axial diffusion effectively eliminated.

•Restored convection–reaction dominated transport.

•Numerically recovered plug-flow-like behavior.

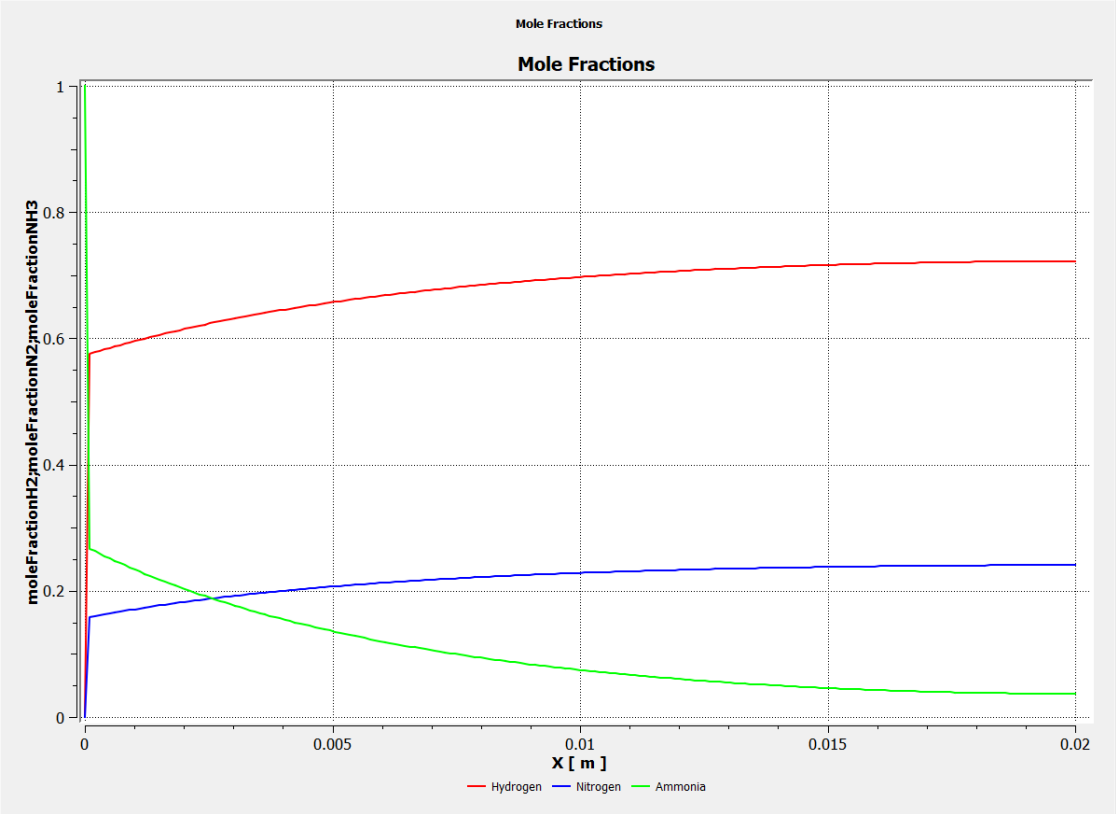
Combustor Zone

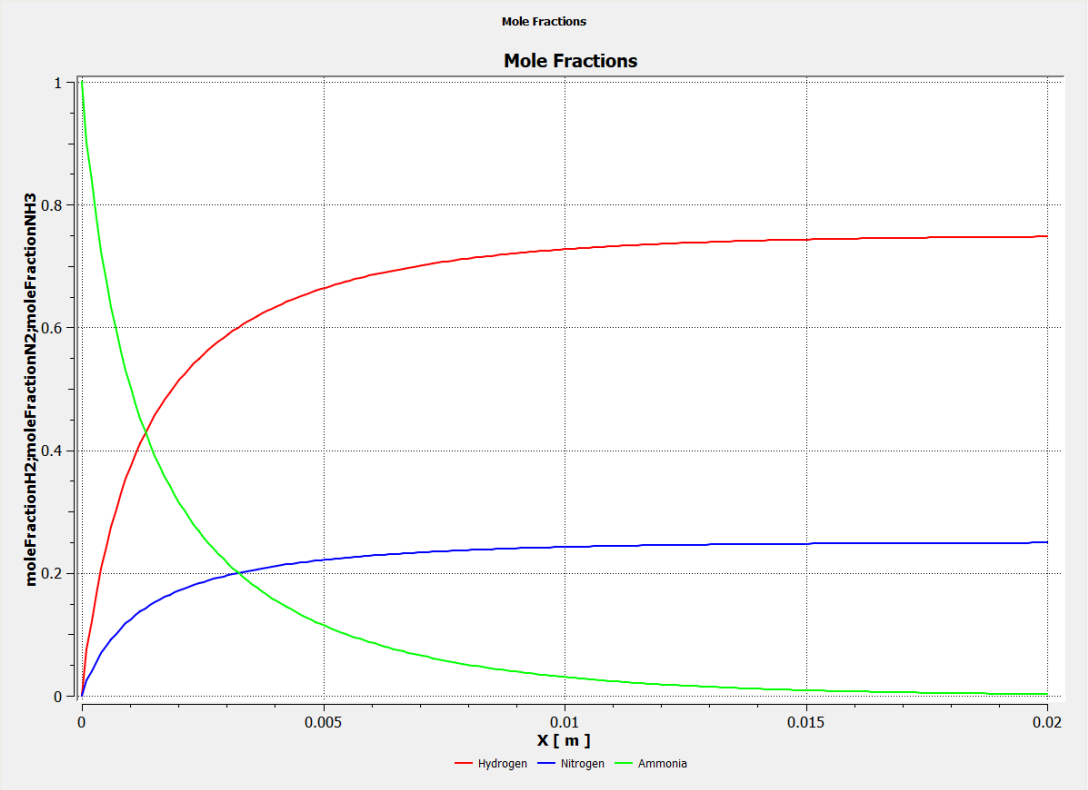
•Species diffusivity computed using Fluent's Chapman–Enskog kinetic-theory formulation.

•Preserved physically realistic diffusion flame structure.

Outcome

- Successfully reconciled two fundamentally different transport regimes within a single CFD framework.
- Simultaneously preserved:
  - plug-flow decomposition physics
  - diffusion-controlled combustion physics
- Enabled stable fully coupled annular cracker–combustor simulations.





Fully Coupled Simulation

Numerical Challenge
Decomposer requires **plug-flow (negligible diffusion)**
Combustor requires **diffusion for flame modeling**
Standard CFD cannot satisfy both simultaneously

Solution: Zone-Specific Diffusivity (UDF)
Decomposer:
    Mass diffusivity → **artificially low** (suppress axial diffusion)
Combustor:
    Mass diffusivity → **Chapman–Enskog model**
Implemented via **zone-based UDF**

Recirculation via UDF
Outlet → inlet coupling implemented using UDFs
Enables **closed-loop thermal interaction**

The decomposer outlet and the fuel inlet for the combustor is physically coupled via a conical connection.

Recirculation via UDF

Outlet ⟶ inlet coupling implemented using UDFs

Enables **closed-loop thermal interaction**

Coupling Strategy

Initialization:

Inlet values from standalone simulations

During simulation the area-weighted outlet properties computed:

- Temperature
- Species composition
- C_p ,molecular weight ,etc.

Recirculated Mass Flow: $\dot{m}_{recirc} = \phi \cdot \dot{m}_{outlet}$

Updated values applied at **flue inlet.**

Outcome

- Self-consistent coupling achieved
- Decomposer maintained at ≈ **893 K**
- Stable convergence obtained

System behaves as a **dynamically coupled thermal loop**, not independent components

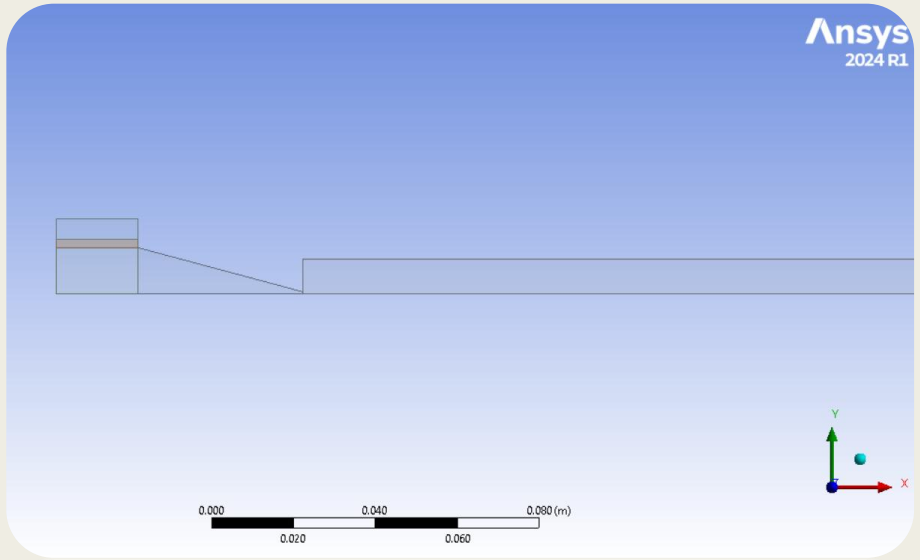
The UDF effectively closes the loop between the combustor and decomposer, allowing the system to converge to its own thermally consistent operating point.

Geometry & Meshing Strategy

Domain Configuration

Decomposer and combustor connected via a **conical transition section**

Ensures smooth flow transfer between regions



Meshing Approach

Structured quadrilateral (face meshing) used in:

- Decomposer
- Combustor

Provides:

- Accurate resolution of **axial gradients**
- Improved **numerical stability** in reacting regions

Conical section (transition region) meshed using **quad-dominant elements**

Enables:

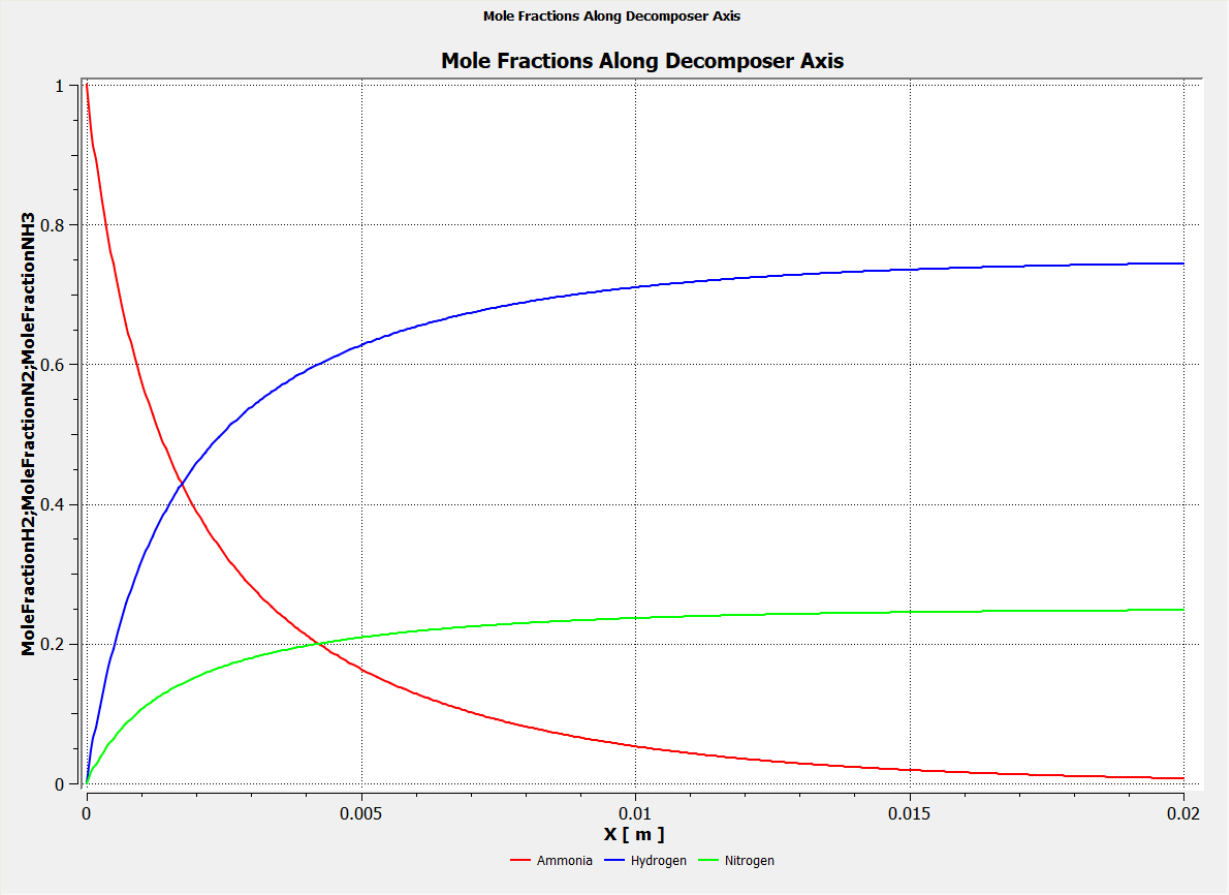
- Smooth geometric transition
- Preservation of mesh quality

Key Advantage

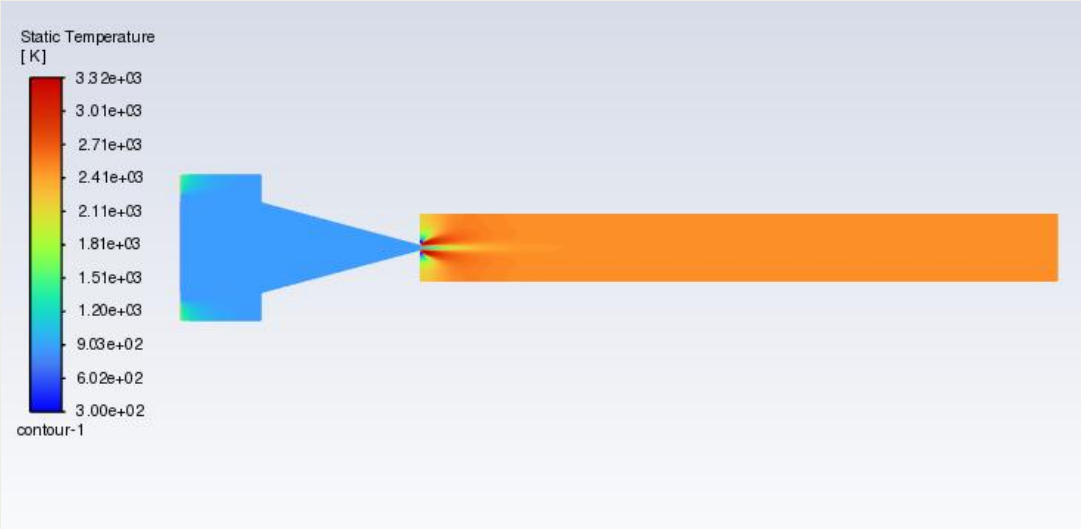
Hybrid meshing ensures:

- Consistent resolution across domains
- Flexibility to handle geometric complexity

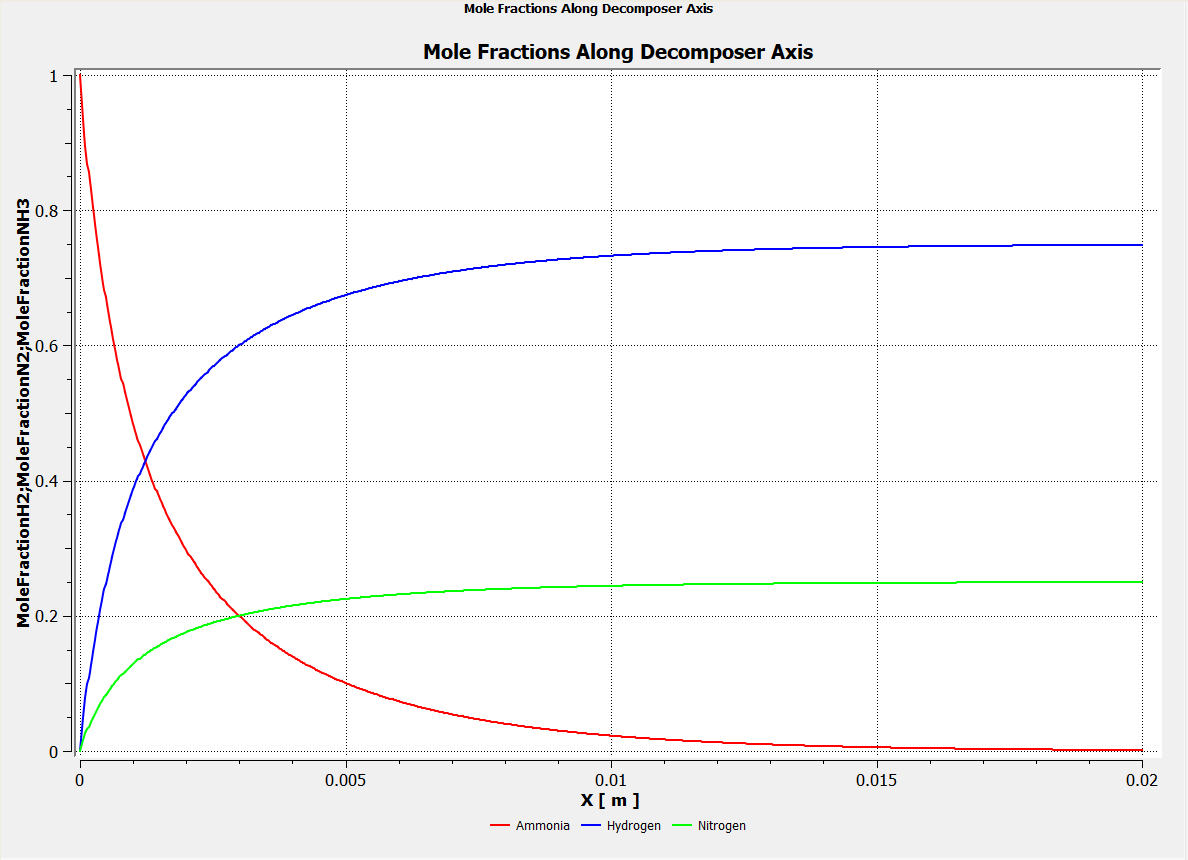
Results:



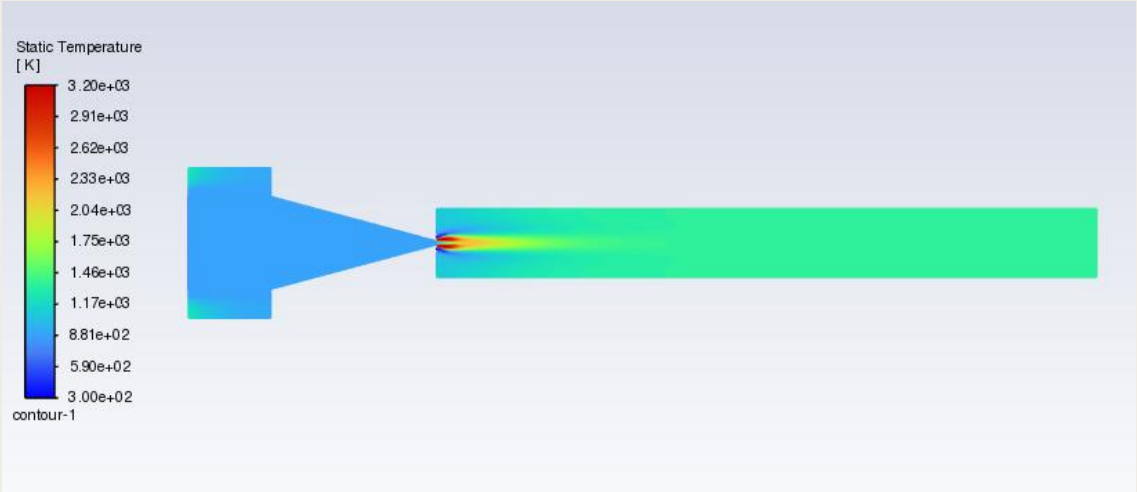
Decomposer Domain Temperature-885K



Outlet Temperature-2478.2935K
26.35% of Flue Gas Recirculated



Decomposer Domain Temperature-
897K



Outlet Temperature-1354.0023K
50% of Flue Gas Recirculated

Conclusion

Key Outcomes

- Validated ammonia decomposer model → **~99.7% conversion**
- Scaled reactor while preserving **plug-flow behavior**
- Designed stable **hydrogen–air combustor**

Thermal Coupling

- Developed **annular decomposer configuration**
- Recirculation ratio ϕpredicted via **MATLAB energy balance**
- Good agreement with CFD results

Fully Coupled System

- Implemented **UDF-based closed-loop coupling**
- Resolved conflicting physics:
  - Plug-flow vs diffusion
- Achieved **self-sustained operation (~893 K)**

Key Insight

- System governed by **energy balance ($\dot{m}$, φ)**
- Not limited by geometric heat transfer

Final Outcome

- Demonstrated a **self-sustaining ammonia decomposer–combustor system**
- Established a **robust multi-physics CFD framework**